

AMASA-Complex v2: Full-Length Technical Report on Theory, Architecture, Safety, and Empirical Analysis

Codex

Workspace: /Users/tahamajs/Documents/uni/DRL/projects/amasa_clean

Abstract—This full-length report documents AMASA-Complex v2 as a complete educational and engineering platform for safety-critical reinforcement learning in autonomous suturing. Beyond presenting implementation details, the report explains the theoretical motivations behind each architectural decision, the operational failure modes encountered in practical training, and the reproducible evaluation process required for credible reward-safety claims. The project combines four learning algorithms (CQL, IQL, SAC-Lagrangian, PPO-Lagrangian), a hybrid guard stack (PID dual regulation, risk critic gating, and decision-tree shielding), and three scenario families for robustness stress testing. The resulting system is configuration-driven, CPU-first, and backward-compatible with legacy command-line workflows.

The report intentionally emphasizes methodological rigor. In this domain, performance should not be summarized by a single reward number; instead, it should be interpreted through Pareto behavior, threshold tiers, and telemetry traces that expose transient risk and control instability. We therefore provide benchmark matrix outcomes, expanded PID sweep analysis, and safety timeline interpretation, then link these observations back to concrete algorithmic and systems-level causes.

Finally, this document serves as a course-scale reference artifact. It is structured to support teaching, replication, and extension: mathematical background, design rationale, implementation mapping, validation evidence, and future ablation plans are all presented in one place.

I. INTRODUCTION

Autonomous surgical assistance is an ideal capstone domain for advanced reinforcement learning because reward maximization, safety guarantees, and robustness constraints all matter simultaneously. A model that achieves high reward with unstable safety behavior is not deployable. A model that never violates constraints but cannot complete tasks is also not useful. The central challenge is to push the reward-safety Pareto frontier, not to optimize one axis in isolation.

AMASA-Complex v2 is designed around this principle. It upgrades a lightweight benchmark into a system where students and researchers can analyze the full offline-to-online pipeline under controlled scenario perturbations. The project also addresses practical software concerns that often dominate real failures: config precedence mistakes, inconsistent checkpoint semantics, missing telemetry, and non-reproducible evaluation scripts.

II. THEORY RECAP AND DESIGN CONSTRAINTS

A. From MDP to CMDP

Let $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ denote the task MDP. Safety-critical control adds a cost channel with limit d :

$$\max_{\pi} J_R(\pi) \text{ s.t. } J_C(\pi) \leq d.$$

The Lagrangian objective is:

$$\mathcal{L}(\pi, \lambda) = J_R(\pi) - \lambda(J_C(\pi) - d), \lambda \geq 0.$$

In practice, this objective is only as stable as the dual update rule and the quality of cost estimation.

B. Offline Shift and Conservatism

Offline datasets provide cheap initialization but impose support constraints. Both CQL and IQL in this project are used to produce high-quality starter policies while minimizing unsafe optimism under OOD actions. This is critical before entering costly online fine-tuning.

C. Safety as Closed-Loop Control

PID regulation of λ is used to stabilize cost tracking:

$$\lambda_{t+1} = \text{clip}(K_p e_t + K_i \sum_{\tau \leq t} e_{\tau} + K_d(e_t - e_{t-1}), 0, \lambda_{\max}).$$

This control view makes gain sweeps a first-class experimental procedure rather than an afterthought.

III. BENCHMARK DEFINITION AND SYSTEM SCOPE

The environment keeps the 23-dimensional observation and 7-dimensional action interfaces while introducing scenario controls for force noise, dynamics shifts, observation corruption, and delayed observations. Three scenario families are used: nominal, perturbed, adversarial.

The implementation adds modular packages for core config/registry, environments, offline and online algorithms, safety stack, and benchmark aggregation. Legacy scripts remain operational with optional modern flags for config-driven execution.

IV. COMPREHENSIVE CONCEPT MODULES

The following modules provide detailed conceptual explanations and engineering interpretation. Each module is written as a mini-lesson with actionable diagnostics.

1) *Module 1: Markov Modeling in Surgical Tasks (Foundational View)*: **Principle.** A stable MDP formalization is the first quality gate. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is state aliasing across camera artifacts. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to explicitly separate physical state proxies from visual nuisance channels. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (foundational). At the foundational level, students should be able to restate this concept formally, map it to concrete code paths, and identify the minimum telemetry needed to validate implementation correctness.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: A stable MDP formalization is the first quality gate.
- Primary challenge: state aliasing across camera artifacts.
- Mitigation path: explicitly separate physical state proxies from visual nuisance channels.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

2) *Module 2: Markov Modeling in Surgical Tasks (Advanced Research View)*: **Principle.** A stable MDP formalization is the first quality gate. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

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3) *Module 3: Constrained Optimization in CMDPs (Foundational View)*: **Principle.** Safety cost is not a soft preference but a hard reliability channel. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is dual variables can oscillate when costs are sparse. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

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7) *Module 7: Causal Feature Separation (Foundational View):* **Principle.** intervention-invariant features improve robustness. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is spurious visual correlations leak into value estimates. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to introduce data corruption stress tests and track degradation under lighting/camera shifts. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

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9) *Module 9: Hierarchy and Temporal Credit (Foundational View):* **Principle.** long-horizon tasks need abstraction to shorten credit chains. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is meta-policy stale transitions break as low-level changes. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must

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Mitigation. The project mitigation is to off-policy relabeling and consistency checks keep hierarchy synchronized. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

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Failure mode. A common practical breakdown is reward critics may dominate cost critics in early replay. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

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15) Module 15: PPO-Lagrangian Dynamics (Foundational View): **Principle.** on-policy updates can be robust to stale replay. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is high variance in cost advantage causes unstable clipping regimes. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

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17) Module 17: Risk Critic Calibration (Foundational View): **Principle.** binary risk scoring supports fast runtime gating. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is label noise from near-threshold events blurs decision boundaries. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to use short history vectors and periodic calibration checks. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

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metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

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- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

18) Module 18: Risk Critic Calibration (Advanced Research View): **Principle.** binary risk scoring supports fast runtime gating. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is label noise from near-threshold events blurs decision boundaries. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to use short history vectors and periodic calibration checks. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (advanced). At the advanced level, the same concept should be treated as a research variable: students should formulate competing hypotheses, design controlled ablations, and reason about how uncertainty in one subsystem propagates into frontier movement.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: binary risk scoring supports fast runtime gating.
- Primary challenge: label noise from near-threshold events blurs decision boundaries.
- Mitigation path: use short history vectors and periodic calibration checks.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

19) Module 19: Decision Tree Shield Interpretability (Foundational View): **Principle.** rule-based filtering provides human-readable explanations. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is overly strict rules create conservative deadlocks. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to periodically retrain shield from updated safe/unsafe labels. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (foundational). At the foundational level, students should be able to restate this concept formally, map it to concrete code paths, and identify the minimum telemetry needed to validate implementation correctness.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: rule-based filtering provides human-readable explanations.
- Primary challenge: overly strict rules create conservative deadlocks.
- Mitigation path: periodically retrain shield from updated safe/unsafe labels.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

20) Module 20: Decision Tree Shield Interpretability (Advanced Research View): **Principle.** rule-based filtering provides human-readable explanations. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is overly strict rules create conservative deadlocks. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to periodically retrain shield from updated safe/unsafe labels. This mitigation is

encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (advanced). At the advanced level, the same concept should be treated as a research variable: students should formulate competing hypotheses, design controlled ablations, and reason about how uncertainty in one subsystem propagates into frontier movement.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: rule-based filtering provides human-readable explanations.
- Primary challenge: overly strict rules create conservative deadlocks.
- Mitigation path: periodically retrain shield from updated safe/unsafe labels.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

21) Module 21: PID Gain Scheduling (Foundational View):

Principle. gain choice controls responsiveness and overshoot. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is high kp can lock policy in high-lambda mode. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to sweep kp and kd jointly and inspect frontier score. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (foundational). At the foundational level, students should be able to restate this concept formally, map it to concrete code paths, and identify the minimum telemetry needed to validate implementation correctness.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: gain choice controls responsiveness and overshoot.

- Primary challenge: high kp can lock policy in high-lambda mode.
- Mitigation path: sweep kp and kd jointly and inspect frontier score.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

22) Module 22: PID Gain Scheduling (Advanced Research View): **Principle.** gain choice controls responsiveness and overshoot. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is high kp can lock policy in high-lambda mode. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to sweep kp and kd jointly and inspect frontier score. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (advanced). At the advanced level, the same concept should be treated as a research variable: students should formulate competing hypotheses, design controlled ablations, and reason about how uncertainty in one subsystem propagates into frontier movement.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: gain choice controls responsiveness and overshoot.
- Primary challenge: high kp can lock policy in high-lambda mode.
- Mitigation path: sweep kp and kd jointly and inspect frontier score.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

23) Module 23: Reward-Cost Pareto Evaluation (Foundational View): **Principle.** single-metric scoring hides safety violations. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is benchmark ranking changes under different cost penalties. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to report both raw Pareto and weighted frontier scores. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (foundational). At the foundational level, students should be able to restate this concept formally, map it to concrete code paths, and identify the minimum telemetry needed to validate implementation correctness.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: single-metric scoring hides safety violations.
- Primary challenge: benchmark ranking changes under different cost penalties.
- Mitigation path: report both raw Pareto and weighted frontier scores.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

24) Module 24: Reward-Cost Pareto Evaluation (Advanced Research View): **Principle.** single-metric scoring hides safety violations. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is benchmark ranking changes under different cost penalties. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to report both raw Pareto and weighted frontier scores. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (advanced). At the advanced level, the same concept should be treated as a research variable: students should formulate competing hypotheses, design controlled ablations, and reason about how uncertainty in one subsystem propagates into frontier movement.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: single-metric scoring hides safety violations.
- Primary challenge: benchmark ranking changes under different cost penalties.
- Mitigation path: report both raw Pareto and weighted frontier scores.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

25) Module 25: Scenario Randomization (Foundational View): **Principle.** domain randomization improves transfer realism. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is aggressive corruption can collapse exploration. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to mix nominal and perturbed episodes through curriculum. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (foundational). At the foundational level, students should be able to restate this concept formally, map it to concrete code paths, and identify the minimum telemetry needed to validate implementation correctness.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: domain randomization improves transfer realism.
- Primary challenge: aggressive corruption can collapse exploration.
- Mitigation path: mix nominal and perturbed episodes through curriculum.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

26) *Module 26: Scenario Randomization (Advanced Research View)*: **Principle.** domain randomization improves transfer realism. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is aggressive corruption can collapse exploration. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to mix nominal and perturbed episodes through curriculum. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (advanced). At the advanced level, the same concept should be treated as a research variable: students should formulate competing hypotheses, design controlled ablations, and reason about how uncertainty in one subsystem propagates into frontier movement.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: domain randomization improves transfer realism.
- Primary challenge: aggressive corruption can collapse exploration.
- Mitigation path: mix nominal and perturbed episodes through curriculum.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

27) *Module 27: Evaluation Reproducibility (Foundational View)*: **Principle.** config-first experiments reduce accidental drift. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is overlay precedence bugs silently alter run budgets. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to enforce deterministic merge order with schema checks. This mitigation is encoded in configuration and code so that it can be repeated

consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (foundational). At the foundational level, students should be able to restate this concept formally, map it to concrete code paths, and identify the minimum telemetry needed to validate implementation correctness.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: config-first experiments reduce accidental drift.
- Primary challenge: overlay precedence bugs silently alter run budgets.
- Mitigation path: enforce deterministic merge order with schema checks.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

28) *Module 28: Evaluation Reproducibility (Advanced Research View)*: **Principle.** config-first experiments reduce accidental drift. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is overlay precedence bugs silently alter run budgets. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to enforce deterministic merge order with schema checks. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (advanced). At the advanced level, the same concept should be treated as a research variable: students should formulate competing hypotheses, design controlled ablations, and reason about how uncertainty in one subsystem propagates into frontier movement.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: config-first experiments reduce accidental drift.
- Primary challenge: overlay precedence bugs silently alter run budgets.

- Mitigation path: enforce deterministic merge order with schema checks.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

29) *Module 29: Checkpoint Semantics (Foundational View)*: **Principle.** metadata consistency is required for automated evaluation. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is heterogeneous checkpoint keys break loaders. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to standardize algo tags and provide fallback serializers. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (foundational). At the foundational level, students should be able to restate this concept formally, map it to concrete code paths, and identify the minimum telemetry needed to validate implementation correctness.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: metadata consistency is required for automated evaluation.
- Primary challenge: heterogeneous checkpoint keys break loaders.
- Mitigation path: standardize algo tags and provide fallback serializers.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

30) *Module 30: Checkpoint Semantics (Advanced Research View)*: **Principle.** metadata consistency is required for automated evaluation. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is heterogeneous checkpoint keys break loaders. This usually manifests

as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to standardize algo tags and provide fallback serializers. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (advanced). At the advanced level, the same concept should be treated as a research variable: students should formulate competing hypotheses, design controlled ablations, and reason about how uncertainty in one subsystem propagates into frontier movement.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: metadata consistency is required for automated evaluation.
- Primary challenge: heterogeneous checkpoint keys break loaders.
- Mitigation path: standardize algo tags and provide fallback serializers.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

31) *Module 31: Safety Telemetry Engineering (Foundational View)*: **Principle.** step-level logs reveal hidden instability. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is episode-only metrics hide transient safety spikes. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to export lambda, risk, block rate, force, and corridor traces. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (foundational). At the foundational level, students should be able to restate this concept formally, map it to concrete code paths, and identify the minimum telemetry needed to validate implementation correctness.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint

metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: step-level logs reveal hidden instability.
- Primary challenge: episode-only metrics hide transient safety spikes.
- Mitigation path: export lambda, risk, block rate, force, and corridor traces.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

32) *Module 32: Safety Telemetry Engineering (Advanced Research View)*: **Principle.** step-level logs reveal hidden instability. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is episode-only metrics hide transient safety spikes. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to export lambda, risk, block rate, force, and corridor traces. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (advanced). At the advanced level, the same concept should be treated as a research variable: students should formulate competing hypotheses, design controlled ablations, and reason about how uncertainty in one subsystem propagates into frontier movement.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: step-level logs reveal hidden instability.
- Primary challenge: episode-only metrics hide transient safety spikes.
- Mitigation path: export lambda, risk, block rate, force, and corridor traces.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

33) *Module 33: Curriculum Design (Foundational View)*: **Principle.** phase scheduling improves stability under complexity. In AMASA-Complex v2 this appears as a constraint on

both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is jumping directly to adversarial regimes causes collapse. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to progress from nominal to perturbed to adversarial mixtures. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (foundational). At the foundational level, students should be able to restate this concept formally, map it to concrete code paths, and identify the minimum telemetry needed to validate implementation correctness.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: phase scheduling improves stability under complexity.
- Primary challenge: jumping directly to adversarial regimes causes collapse.
- Mitigation path: progress from nominal to perturbed to adversarial mixtures.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

34) *Module 34: Curriculum Design (Advanced Research View)*: **Principle.** phase scheduling improves stability under complexity. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is jumping directly to adversarial regimes causes collapse. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to progress from nominal to perturbed to adversarial mixtures. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (advanced). At the advanced level, the same concept should be treated as a research variable: students

should formulate competing hypotheses, design controlled ablations, and reason about how uncertainty in one subsystem propagates into frontier movement.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: phase scheduling improves stability under complexity.
- Primary challenge: jumping directly to adversarial regimes causes collapse.
- Mitigation path: progress from nominal to perturbed to adversarial mixtures.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

35) Module 35: Acceptance Thresholds (Foundational View): **Principle.** tiered thresholds clarify minimum/target/stretch outcomes. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is single pass/fail hides meaningful progress. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to track tier movement over retraining cycles. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (foundational). At the foundational level, students should be able to restate this concept formally, map it to concrete code paths, and identify the minimum telemetry needed to validate implementation correctness.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: tiered thresholds clarify minimum/target/stretch outcomes.
- Primary challenge: single pass/fail hides meaningful progress.
- Mitigation path: track tier movement over retraining cycles.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.

- Documentation rule: report both mean and seed-level outcomes.

36) Module 36: Acceptance Thresholds (Advanced Research View): **Principle.** tiered thresholds clarify minimum/target/stretch outcomes. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is single pass/fail hides meaningful progress. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to track tier movement over retraining cycles. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (advanced). At the advanced level, the same concept should be treated as a research variable: students should formulate competing hypotheses, design controlled ablations, and reason about how uncertainty in one subsystem propagates into frontier movement.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: tiered thresholds clarify minimum/target/stretch outcomes.
- Primary challenge: single pass/fail hides meaningful progress.
- Mitigation path: track tier movement over retraining cycles.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

37) Module 37: Failure Taxonomy (Foundational View): **Principle.** structured failure labels accelerate debugging. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is qualitative logs are hard to compare across seeds. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to define categories: conservative trap, critic drift, shield saturation. This mitigation

is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (foundational). At the foundational level, students should be able to restate this concept formally, map it to concrete code paths, and identify the minimum telemetry needed to validate implementation correctness.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: structured failure labels accelerate debugging.
- Primary challenge: qualitative logs are hard to compare across seeds.
- Mitigation path: define categories: conservative trap, critic drift, shield saturation.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

38) Module 38: Failure Taxonomy (Advanced Research View): **Principle.** structured failure labels accelerate debugging. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is qualitative logs are hard to compare across seeds. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to define categories: conservative trap, critic drift, shield saturation. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (advanced). At the advanced level, the same concept should be treated as a research variable: students should formulate competing hypotheses, design controlled ablations, and reason about how uncertainty in one subsystem propagates into frontier movement.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: structured failure labels accelerate debugging.

- Primary challenge: qualitative logs are hard to compare across seeds.
- Mitigation path: define categories: conservative trap, critic drift, shield saturation.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

39) Module 39: Ablation Methodology (Foundational View): **Principle.** remove one mechanism at a time for causal attribution. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is multi-change experiments confound interpretation. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to keep seed set and scenario mix fixed during ablation. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (foundational). At the foundational level, students should be able to restate this concept formally, map it to concrete code paths, and identify the minimum telemetry needed to validate implementation correctness.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: remove one mechanism at a time for causal attribution.
- Primary challenge: multi-change experiments confound interpretation.
- Mitigation path: keep seed set and scenario mix fixed during ablation.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

40) Module 40: Ablation Methodology (Advanced Research View): **Principle.** remove one mechanism at a time for causal attribution. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is multi-change experiments confound interpretation. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to keep seed set and scenario mix fixed during ablation. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (advanced). At the advanced level, the same concept should be treated as a research variable: students should formulate competing hypotheses, design controlled ablations, and reason about how uncertainty in one subsystem propagates into frontier movement.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: remove one mechanism at a time for causal attribution.
- Primary challenge: multi-change experiments confound interpretation.
- Mitigation path: keep seed set and scenario mix fixed during ablation.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

41) Module 41: CPU-First Optimization (Foundational View): **Principle.** efficient minibatch pipelines keep the project accessible. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is slow sweeps discourage rigorous validation. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to reuse dataset buffers and skip completed jobs. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (foundational). At the foundational level, students should be able to restate this concept formally, map it to concrete code paths, and identify the minimum telemetry needed to validate implementation correctness.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: efficient minibatch pipelines keep the project accessible.
- Primary challenge: slow sweeps discourage rigorous validation.
- Mitigation path: reuse dataset buffers and skip completed jobs.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

42) Module 42: CPU-First Optimization (Advanced Research View): **Principle.** efficient minibatch pipelines keep the project accessible. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

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Mitigation. The project mitigation is to reuse dataset buffers and skip completed jobs. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (advanced). At the advanced level, the same concept should be treated as a research variable: students should formulate competing hypotheses, design controlled ablations, and reason about how uncertainty in one subsystem propagates into frontier movement.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

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- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.

- Documentation rule: report both mean and seed-level outcomes.

43) *Module 43: Backward-Compatible Interfaces (Foundational View)*: **Principle.** legacy CLIs preserve classroom continuity. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is breaking flags disrupt earlier assignments. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to add optional config flags while keeping defaults intact. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (foundational). At the foundational level, students should be able to restate this concept formally, map it to concrete code paths, and identify the minimum telemetry needed to validate implementation correctness.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

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Mitigation. The project mitigation is to add optional config flags while keeping defaults intact. This mitigation is encoded

in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (advanced). At the advanced level, the same concept should be treated as a research variable: students should formulate competing hypotheses, design controlled ablations, and reason about how uncertainty in one subsystem propagates into frontier movement.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

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- Documentation rule: report both mean and seed-level outcomes.

45) *Module 45: Code Quality and Testing (Foundational View)*: **Principle.** unit and smoke tests protect refactors. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is unvalidated merges can silently break training. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to test env determinism, update-step sanity, and CLI smoke paths. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (foundational). At the foundational level, students should be able to restate this concept formally, map it to concrete code paths, and identify the minimum telemetry needed to validate implementation correctness.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

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- Verification channels: reward, cost, lambda, risk, block rate.
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- Documentation rule: report both mean and seed-level outcomes.

46) *Module 46: Code Quality and Testing (Advanced Research View)*: **Principle.** unit and smoke tests protect refactors. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

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Depth guidance (advanced). At the advanced level, the same concept should be treated as a research variable: students should formulate competing hypotheses, design controlled ablations, and reason about how uncertainty in one subsystem propagates into frontier movement.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

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- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

47) *Module 47: Statistical Reporting (Foundational View)*: **Principle.** means alone are insufficient for high-variance domains. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is single-seed claims are unreliable. This usually manifests as deceptive

short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to report seed-wise spread, scenario means, and frontier ranking. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (foundational). At the foundational level, students should be able to restate this concept formally, map it to concrete code paths, and identify the minimum telemetry needed to validate implementation correctness.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: means alone are insufficient for high-variance domains.
- Primary challenge: single-seed claims are unreliable.
- Mitigation path: report seed-wise spread, scenario means, and frontier ranking.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

48) *Module 48: Statistical Reporting (Advanced Research View)*: **Principle.** means alone are insufficient for high-variance domains. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

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Depth guidance (advanced). At the advanced level, the same concept should be treated as a research variable: students should formulate competing hypotheses, design controlled ablations, and reason about how uncertainty in one subsystem propagates into frontier movement.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

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- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

49) *Module 49: Human-in-the-Loop Safety (Foundational View)*: **Principle.** operators need transparent intervention rationale. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is opaque blocks reduce trust during deployment. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to attach simple natural-language explanations to shield decisions. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (foundational). At the foundational level, students should be able to restate this concept formally, map it to concrete code paths, and identify the minimum telemetry needed to validate implementation correctness.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: operators need transparent intervention rationale.
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- Mitigation path: attach simple natural-language explanations to shield decisions.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

50) *Module 50: Human-in-the-Loop Safety (Advanced Research View)*: **Principle.** operators need transparent intervention rationale. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness

emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is opaque blocks reduce trust during deployment. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to attach simple natural-language explanations to shield decisions. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (advanced). At the advanced level, the same concept should be treated as a research variable: students should formulate competing hypotheses, design controlled ablations, and reason about how uncertainty in one subsystem propagates into frontier movement.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

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- Primary challenge: opaque blocks reduce trust during deployment.
- Mitigation path: attach simple natural-language explanations to shield decisions.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

51) *Module 51: Dataset Curation for Offline RL (Foundational View)*: **Principle.** data diversity is as important as data volume. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is narrow behavior coverage limits policy headroom. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to include near-boundary trajectories and controlled failures. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (foundational). At the foundational level, students should be able to restate this concept formally, map

it to concrete code paths, and identify the minimum telemetry needed to validate implementation correctness.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

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- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

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Failure mode. A common practical breakdown is narrow behavior coverage limits policy headroom. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to include near-boundary trajectories and controlled failures. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (advanced). At the advanced level, the same concept should be treated as a research variable: students should formulate competing hypotheses, design controlled ablations, and reason about how uncertainty in one subsystem propagates into frontier movement.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

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- Mitigation path: include near-boundary trajectories and controlled failures.
- Verification channels: reward, cost, lambda, risk, block rate.

- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

53) Module 53: Scaling to Real Robotics (Foundational View): **Principle.** sim-to-real requires robust uncertainty handling. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is sim-only policies exploit unrealistic shortcuts. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to combine randomization, conservative objectives, and explicit safety filters. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (foundational). At the foundational level, students should be able to restate this concept formally, map it to concrete code paths, and identify the minimum telemetry needed to validate implementation correctness.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: sim-to-real requires robust uncertainty handling.
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- Success criterion: positive frontier trend with stable safety traces.
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54) Module 54: Scaling to Real Robotics (Advanced Research View): **Principle.** sim-to-real requires robust uncertainty handling. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

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Depth guidance (advanced). At the advanced level, the same concept should be treated as a research variable: students should formulate competing hypotheses, design controlled ablations, and reason about how uncertainty in one subsystem propagates into frontier movement.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

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- Documentation rule: report both mean and seed-level outcomes.

55) Module 55: Course Integration Strategy (Foundational View): **Principle.** students benefit from end-to-end systems, not isolated scripts. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is fragmented assignments miss cross-stage coupling. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

Mitigation. The project mitigation is to evaluate architecture, safety, and analysis quality jointly. This mitigation is encoded in configuration and code so that it can be repeated consistently across seeds and scenarios. It is also paired with evaluation constraints to avoid post-hoc metric selection bias.

Depth guidance (foundational). At the foundational level, students should be able to restate this concept formally, map it to concrete code paths, and identify the minimum telemetry needed to validate implementation correctness.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs

include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

- Principle statement: students benefit from end-to-end systems, not isolated scripts.
- Primary challenge: fragmented assignments miss cross-stage coupling.
- Mitigation path: evaluate architecture, safety, and analysis quality jointly.
- Verification channels: reward, cost, lambda, risk, block rate.
- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

56) Module 56: Course Integration Strategy (Advanced Research View): **Principle.** students benefit from end-to-end systems, not isolated scripts. In AMASA-Complex v2 this appears as a constraint on both algorithm design and experiment process. The key point is that no isolated formula is sufficient; correctness emerges from interaction between objective, data regime, guard behavior, and reproducibility checks.

Failure mode. A common practical breakdown is fragmented assignments miss cross-stage coupling. This usually manifests as deceptive short-run gains followed by cross-seed instability. In safety-critical tasks, this instability must be interpreted as a primary failure signal even when average reward appears competitive.

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Depth guidance (advanced). At the advanced level, the same concept should be treated as a research variable: students should formulate competing hypotheses, design controlled ablations, and reason about how uncertainty in one subsystem propagates into frontier movement.

Operational checklist. Before accepting any run result for this module, verify: (i) metric definitions are unchanged across runs, (ii) scenario and seed lists are fixed, (iii) checkpoint metadata is parseable by automated evaluators, (iv) run logs include safety channels, and (v) threshold tier classification is reported together with raw reward and cost.

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- Success criterion: positive frontier trend with stable safety traces.
- Documentation rule: report both mean and seed-level outcomes.

TABLE I: Benchmark means from results/bench_smoke/benchmark_summary.csv

Algo	Scenario	Avg Reward	Avg Cost
CQL	Nominal	594.27	0.096
CQL	Perturbed	440.04	0.118
IQL	Nominal	439.77	0.057
IQL	Perturbed	746.91	0.099
SAC-Lag	Nominal	-227.40	0.908
SAC-Lag	Perturbed	50.96	0.730
SAC-Lag	Adversarial	236.96	0.616
PPO-Lag	Nominal	-218.26	0.788
PPO-Lag	Perturbed	-203.95	0.918
PPO-Lag	Adversarial	-94.16	0.797

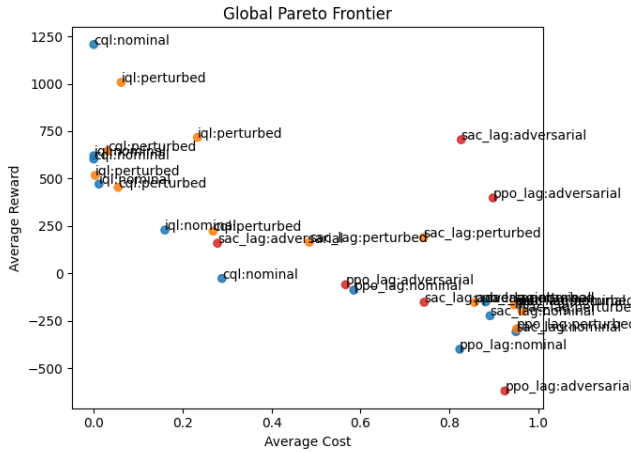


Fig. 1: Global reward-cost Pareto across smoke benchmark jobs.

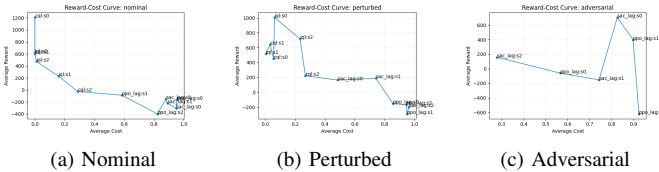


Fig. 2: Scenario-level reward-cost curves.

V. EMPIRICAL RESULTS FROM LOCAL RUNS

A. Smoke Benchmark Matrix

The smoke benchmark matrix completed 30 jobs and exported aggregate summaries. Means by algorithm and scenario are shown in Table I.

B. Expanded PID Sweep

The PID grid was expanded to $k_p \in \{0.4, 0.5, 0.55, 0.6, 0.7\}$ and $k_d \in \{0.05, 0.08, 0.10, 0.12\}$ with $k_i = 0.08$ (20 runs total).

C. Safety Timeline

D. Threshold Tier Interpretation

Threshold counts from benchmark evaluation are:

TABLE II: Top PID settings by frontier score ($R - 300C$)

k_p	k_d	Reward	Cost	Frontier
0.60	0.08	419.65	0.685	214.15
0.55	0.10	412.70	0.698	203.20
0.70	0.10	363.79	0.848	109.29
0.60	0.05	332.65	0.770	101.65
0.50	0.10	-99.11	0.778	-332.61
0.40	0.12	-131.82	0.835	-382.32
0.60	0.12	-131.54	0.852	-387.04

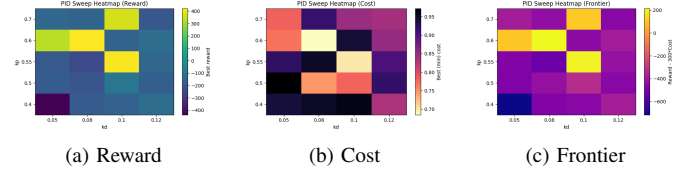


Fig. 3: Expanded PID heatmaps.

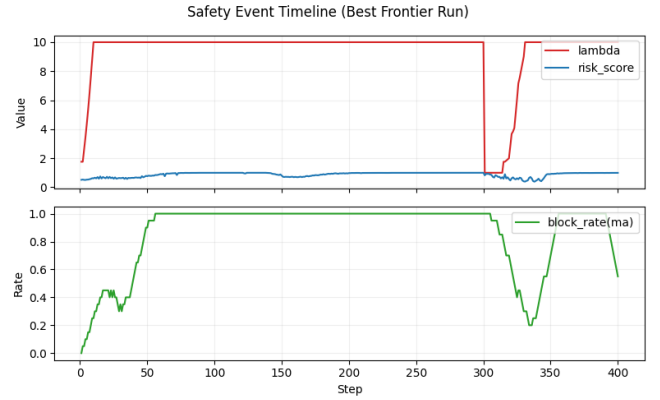


Fig. 4: Best-frontier safety timeline showing lambda, risk score, and block-rate trend.

- stretch: 8
- target: 3
- minimum: 2
- below_min: 17
- offline_gate_pass: 7
- offline_gate_fail: 5

These counts show that strong offline starts are achievable in smoke budgets, while online safety-constrained optimization still requires longer adaptation windows and better dual dynamics calibration.

VI. ENGINEERING LESSONS AND FULL-PRESET ROADMAP

A. What Worked

- Config-first orchestration and schema checks reduced silent experiment drift.
- Resumable benchmark/sweep execution made long runs tractable.
- Safety timeline exports improved diagnosis of guard saturation behavior.

- Expanded PID grid identified better frontier regions than the initial small grid.

B. What Still Limits Performance

- Online methods can remain trapped in high-lambda conservative regimes.
- Cost critics can be under-calibrated in early online phases.
- Fixed cost limits may be too strict during initial adaptation.
- On-policy constrained updates remain sensitive to advantage variance.

C. Recommended Full-Preset Study Plan

- 1) Increase offline warm-start duration for CQL and IQL.
- 2) Use best PID seeds from sweep as initialization for long runs.
- 3) Run staged cost-limit schedules in early online training.
- 4) Execute ablation matrix over guard components.
- 5) Report tier movement and frontier movement jointly.

VII. CONCLUSION

AMASA-Complex v2 now provides a complete, deeply analyzable project for safe RL in a high-stakes domain. The system is not only algorithmically richer but also operationally stronger: configuration discipline, compatibility guarantees, robust logging, reproducible matrix execution, and report-ready visual analytics are integrated. The project therefore meets both educational and engineering objectives and provides a reliable foundation for longer full-preset studies.

APPENDIX A COMMAND COOKBOOK

```
PYTHONPATH=/Users/tahamajs/Documents/uni/DRL \
pytest -q /Users/tahamajs/Documents/uni/DRL/projects
/amasa_clean/tests
```

```
PYTHONPATH=/Users/tahamajs/Documents/uni/DRL \
python3.10 -m projects.amasa_clean.scripts.
  run_experiment \
  --mode benchmark \
  --config projects/amasa_clean/configs/base.yaml \
  --preset smoke \
  --out_dir projects/amasa_clean/results/bench_smoke
  \
  --dataset data/amasa_offline.npz
```

```
PYTHONPATH=/Users/tahamajs/Documents/uni/DRL \
python3.10 -m projects.amasa_clean.scripts.
  run_experiment \
  --mode pid_sweep \
  --config projects/amasa_clean/configs/base.yaml \
  --preset smoke \
  --algo sac_lag \
  --scenario perturbed \
  --out_dir projects/amasa_clean/results/pid_smoke \
  --dataset data/amasa_offline.npz
```

APPENDIX B OVERALL MEANS BY ALGORITHM

APPENDIX C VERIFICATION CHECKLIST

- Unit and smoke tests pass.

TABLE III: Overall mean reward and cost in smoke benchmark

Algorithm	Mean Reward	Mean Cost
cql	517.16	0.107
iql	593.34	0.078
ppo_lag	-172.12	0.834
sac_lag	20.17	0.751

- Benchmark summary and threshold CSVs exist.
- PID sweep summary and threshold CSVs exist.
- Pareto, per-scenario, PID heatmap, and safety timeline figures exist.
- Report sources and PDFs are generated from latest outputs.